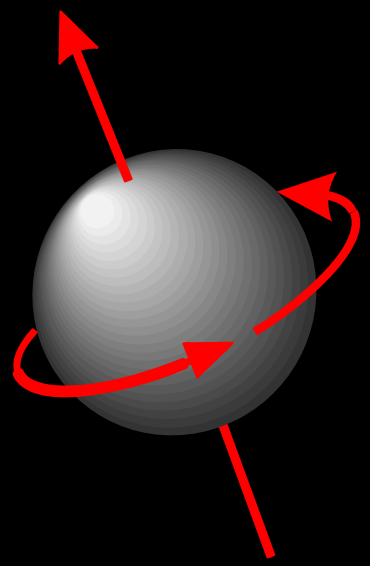
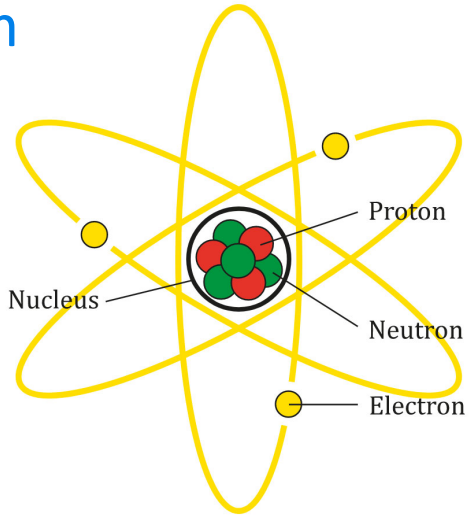
Principle



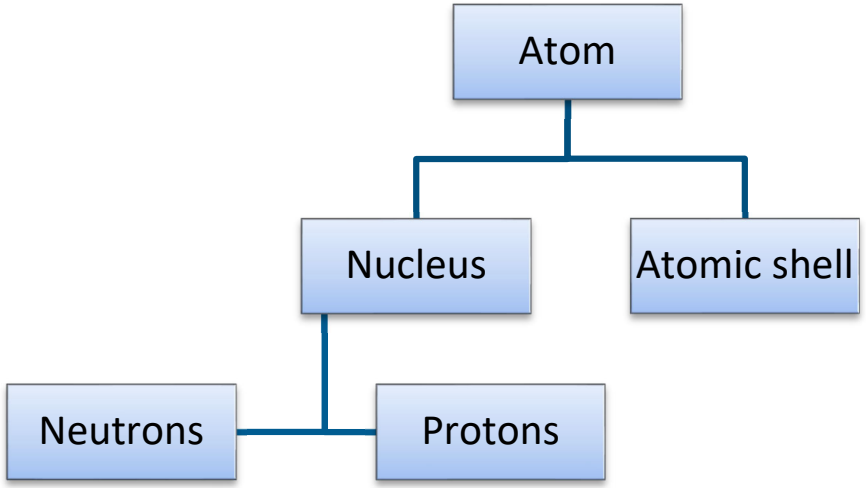
Andreas Hock

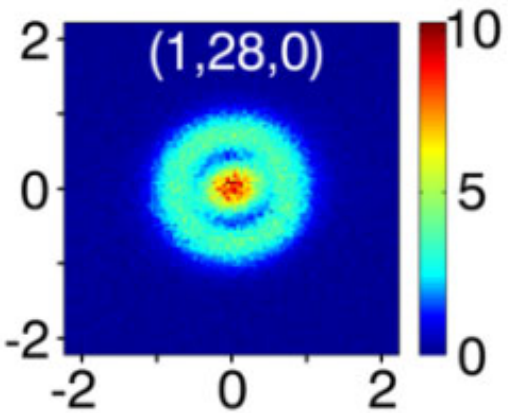
Principle

Atom

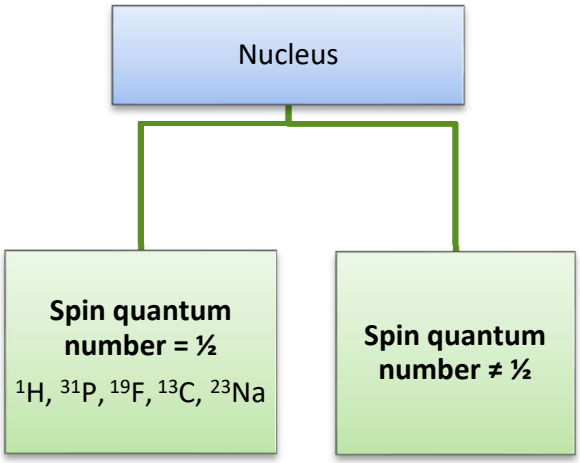


https://upload.wikimedia.org/wikipedia/commons/8/80/Atom_Diagram.svg





Hydrogen Atoms under Magnification: Direct Observation of the Nodal Structure of Stark States, A. S. Stodolna et al. Phys. Rev. Lett. 110, 213001 – Published 20 May 2013



Quantum mechanics

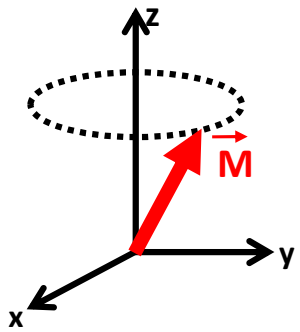
"Expected value" of macroscopic magnetization

Spin quantum number	+1/2	"Spin up" or parallel
	-1/2	"Spin down" or antiparallel

Energy difference $\Delta E = h\nu$

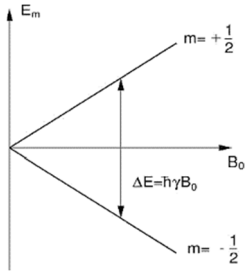
Mathematical formalism of quantum mechanics

Averaging over a large number of nuclei (ensemble mean -> expected value)



Classical vs. Quantum Mechanical description of MRI

Quantum Description of MRI



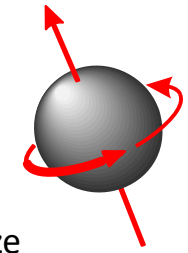
Advantages:

- Detailed and accurate; includes energy quantization
- Explains spin coherence and advanced phenomena
- Essential for novel techniques (e.g., hyperpolarization, spectroscopy)

Disadvantages:

- Complex, abstract, and computationally intensive
- Less intuitive; harder to understand for non-specialists
- Overly detailed for routine clinical use

Classical Description of MRI



Advantages:

- Intuitive and easy to visualize
- Simpler mathematics; accessible to non-specialists
- Effective for hardware design and clinical applications

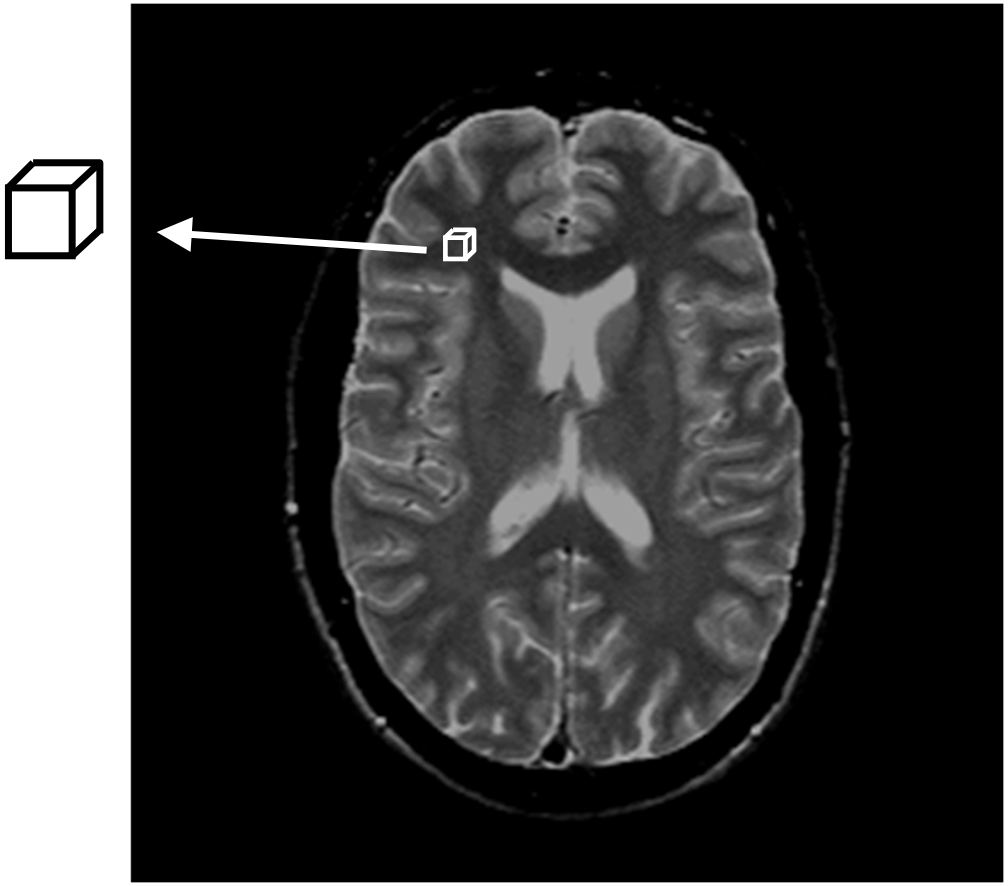
Disadvantages:

- Oversimplifies spin behavior
- Cannot explain e.g. coherence

Principle

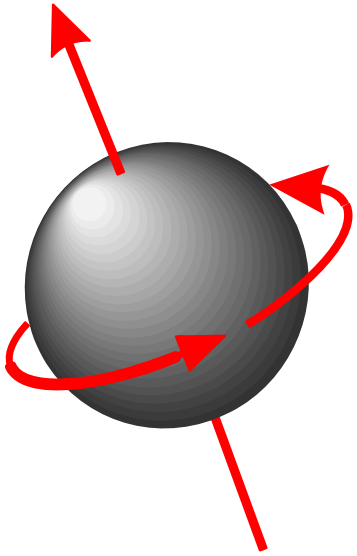
Ensemble of many nuclei

In a mm^3:
10^{19} nuclei!

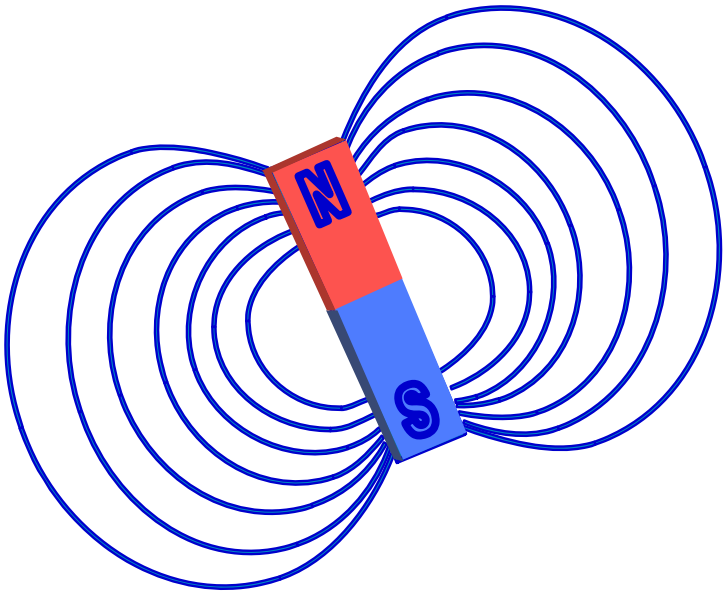


Principle

Rotating nuclei are like tiny magnets



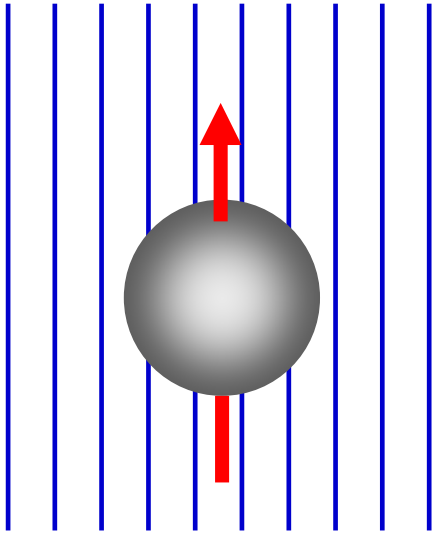
spinning top / gyroscope



Principle

Alignment in the magnetic field

Earth's magnetic field
5.10^{-5} Tesla

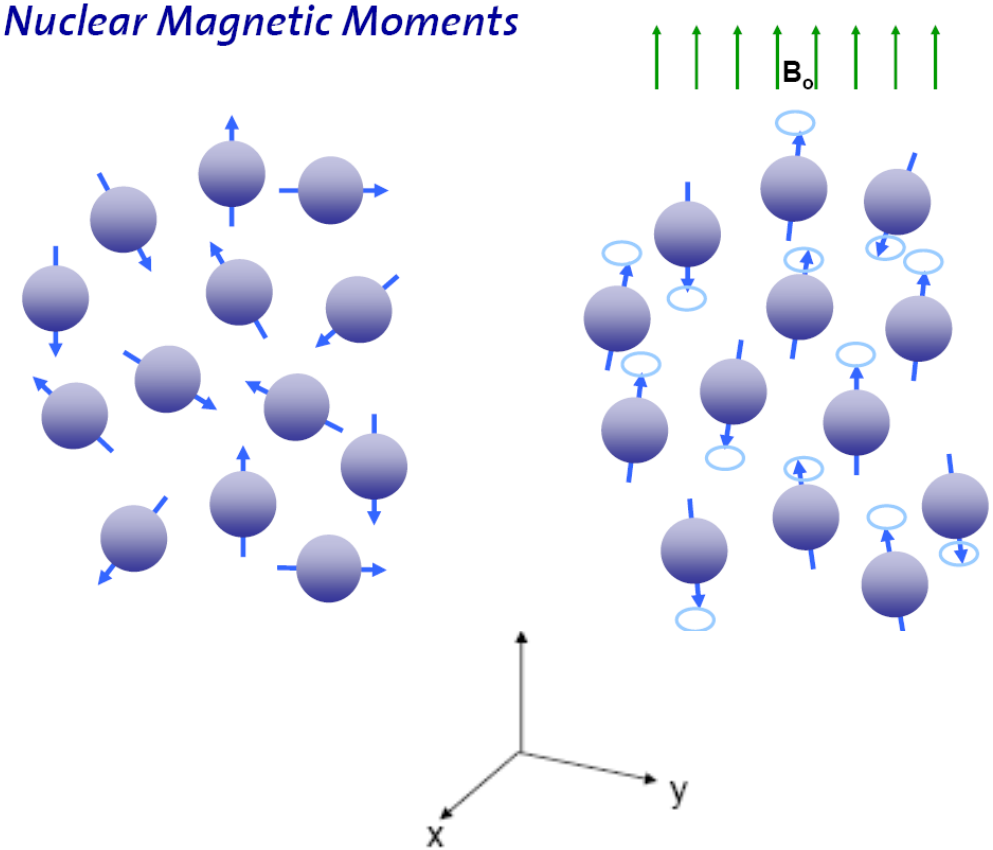


Magnetic field
0.2 – 3.0 Tesla

Principle

Alignment in the magnetic field

Nuclear Magnetic Moments



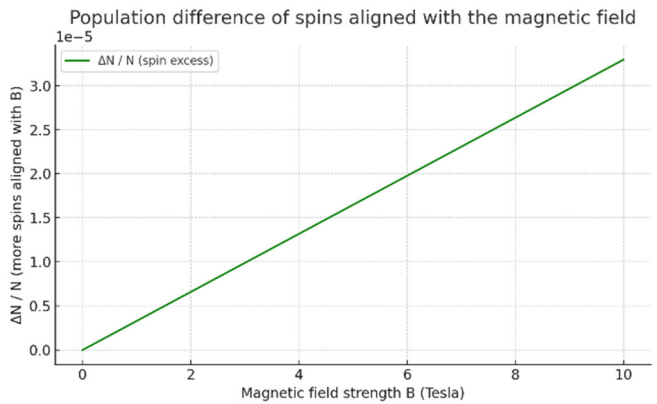
Calculation of Spin Difference

$$\Delta N = N_{\uparrow} - N_{\downarrow} = N \cdot \frac{\gamma \hbar B}{2 k_B T}$$

Where:

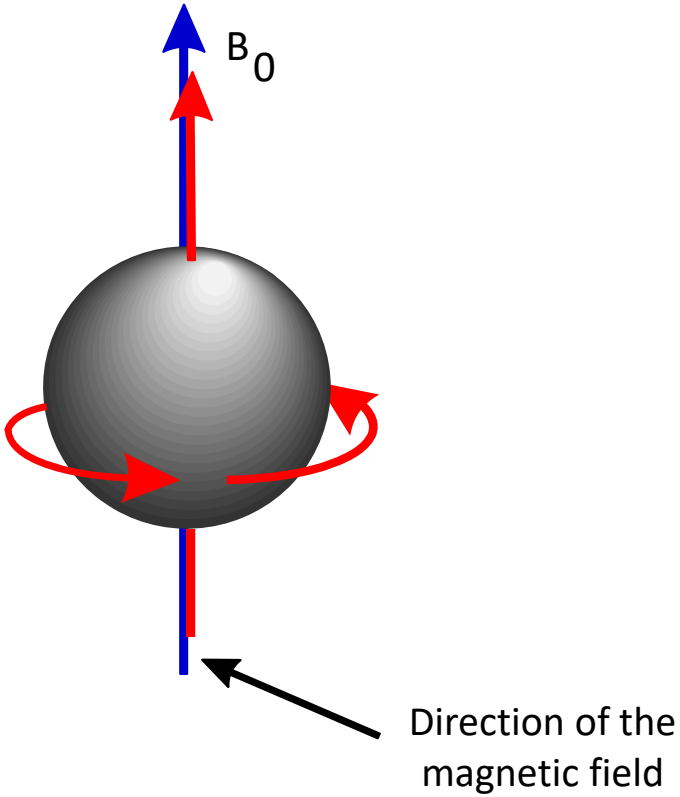
- N is the total number of spins.
- γ is the **gyromagnetic ratio**.
- $\hbar$ is the **reduced Planck constant**.
- B is the magnetic field strength.
- k_B is the **Boltzmann constant**.
- T is the temperature in Kelvin.

At 3T and body temperature:
approximately **1 additional aligned spin per 100,000 spins**



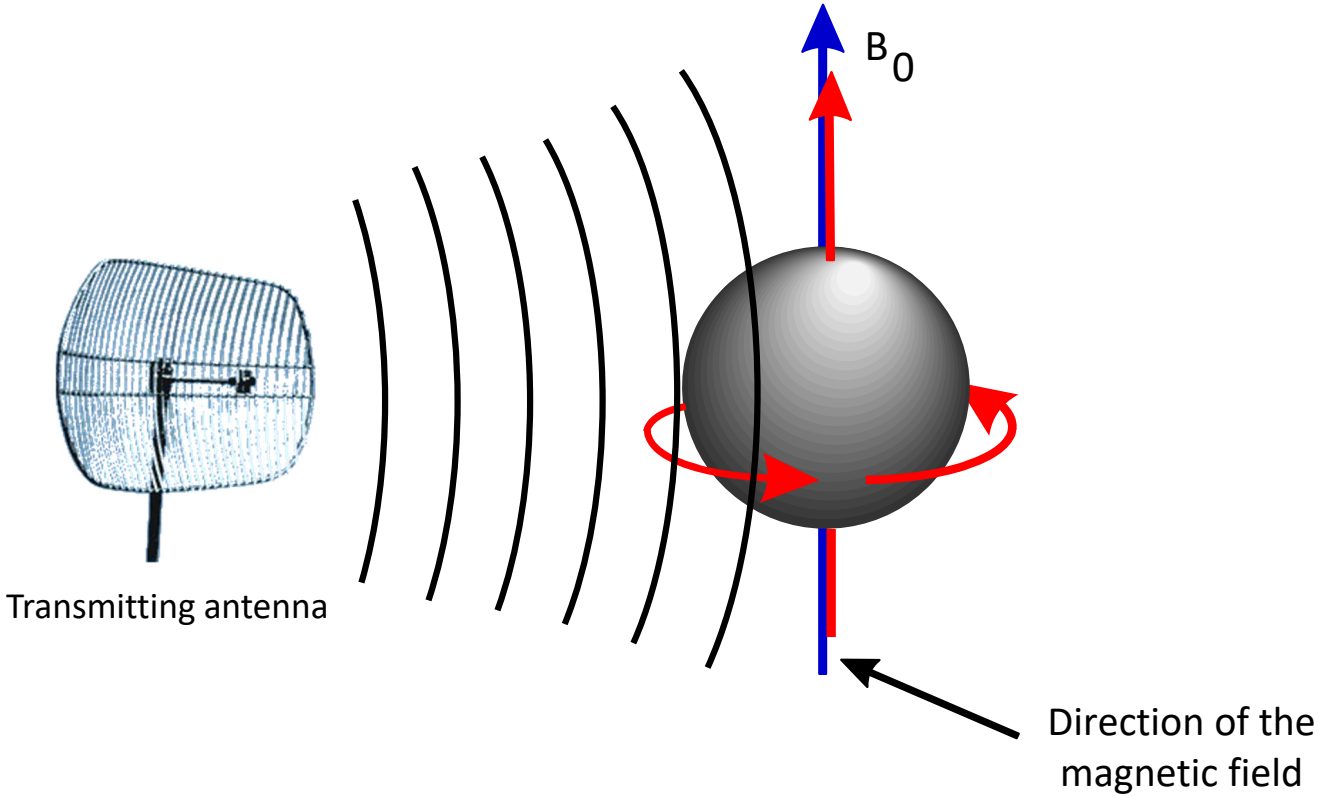
Principle

Alignment in the magnetic field



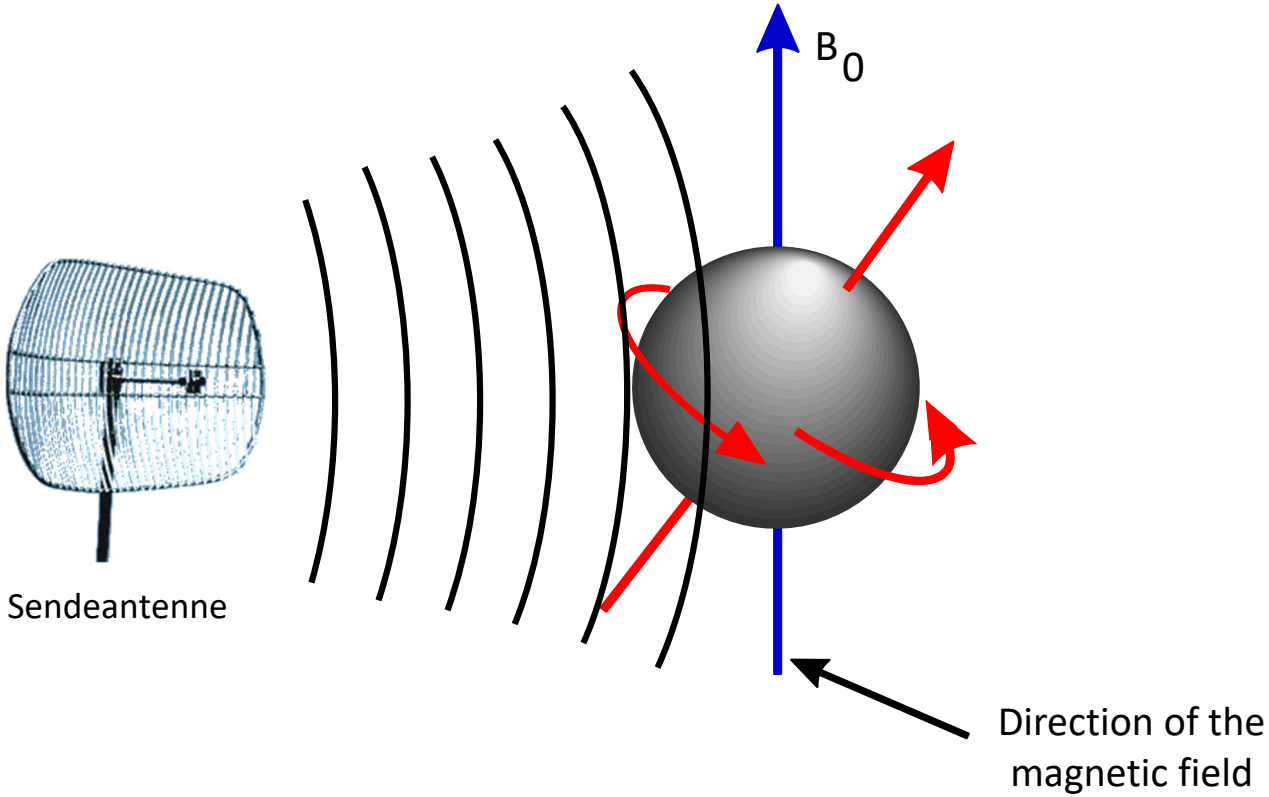
Excitation

with short high-frequency pulse



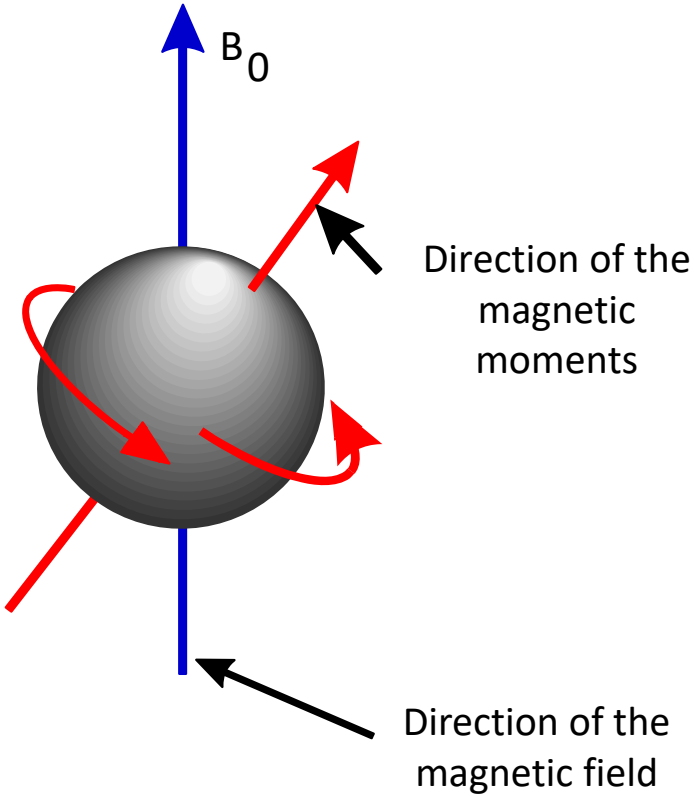
Excitation

with short high-frequency pulse



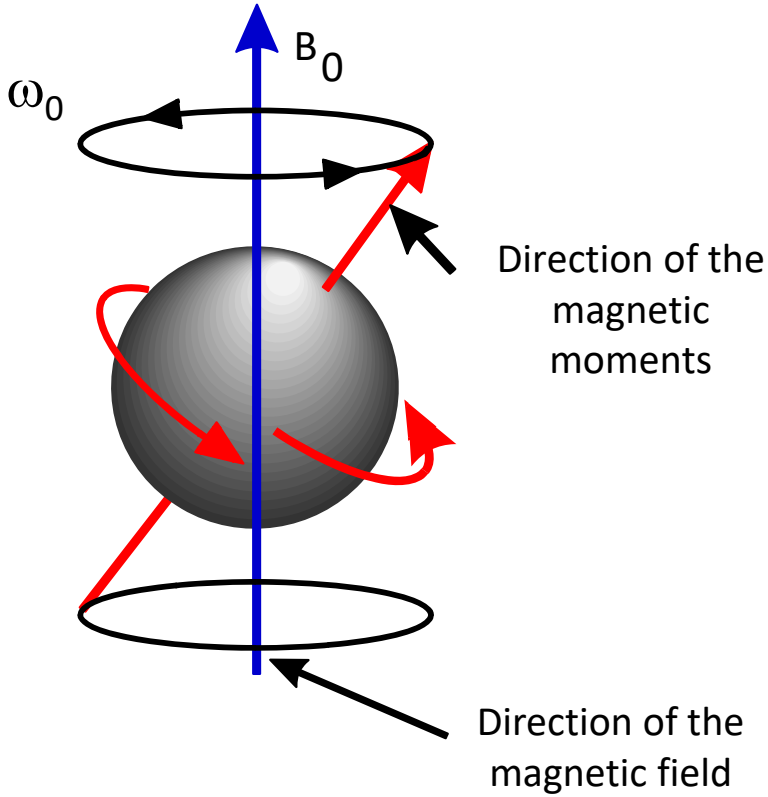
Excitation

Magnetic field and axis of rotation no longer parallel



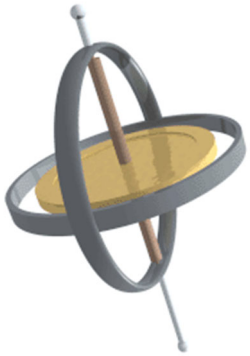
Precession

of the rotating core



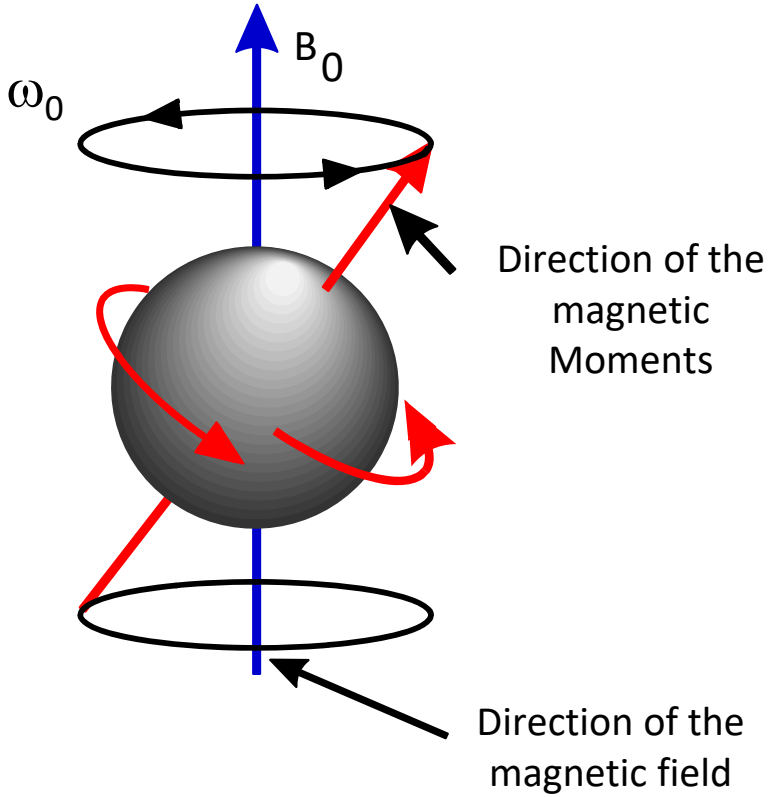
Precession

from the rotating gyro



Precession

of the rotating core



Larmor Frequency

Proportional to magnetic field strength

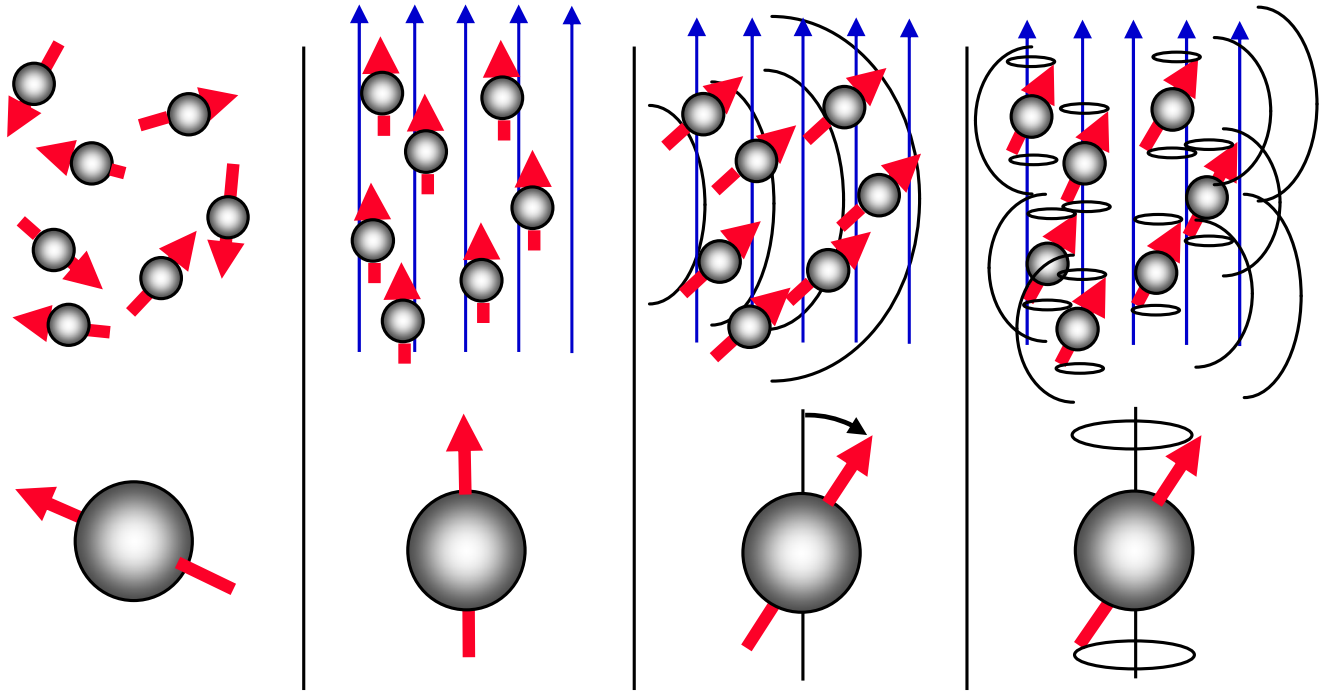
Larmor Frequency: $\omega_0 = \gamma B_0$

γ : gyromagnetic ratio (MHz/T)
B_0 : magnetic flux density (Tesla = A/m)

nuclei	γ (MHz/T)
^{1}H	42.6
^{31}P	17.2
^{19}F	40.0
^{13}C	10.8
^{23}Na	11.2

MR tomography

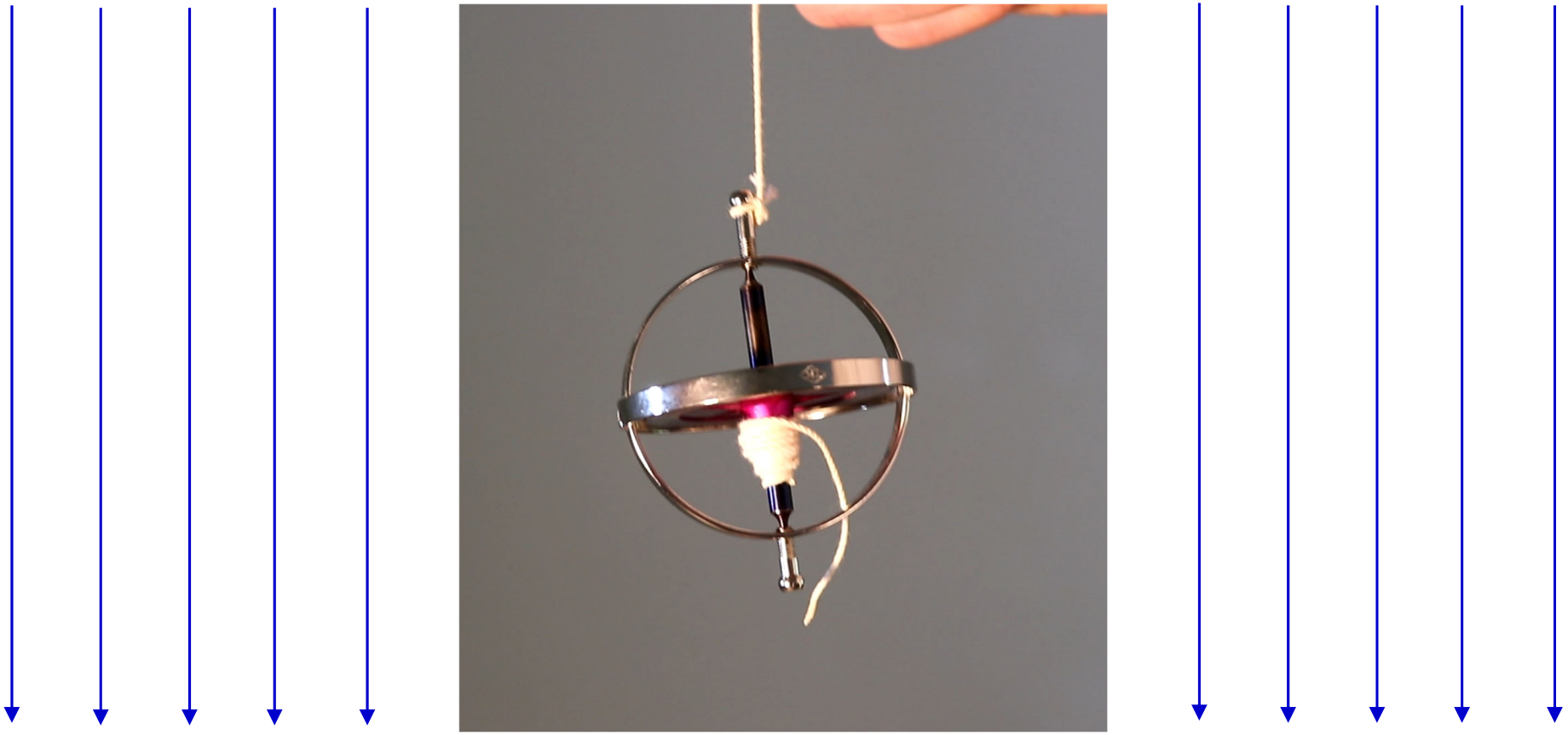
Procedure for MR measurement



Without field | Orientation in the magnetic field | Excitation with RF pulse | Precession

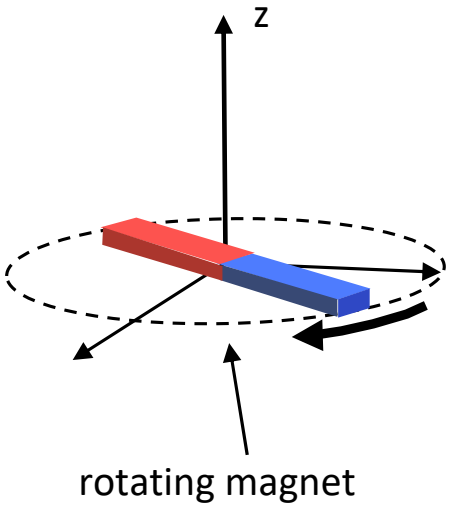
MR tomography

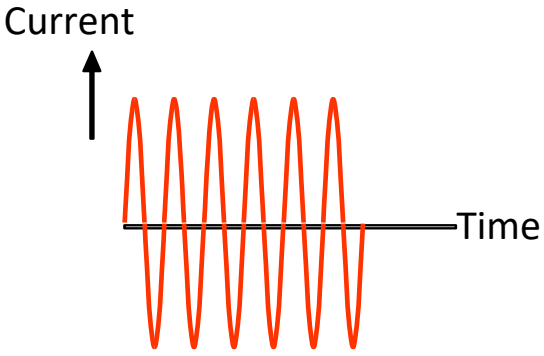
Procedure for MR measurement

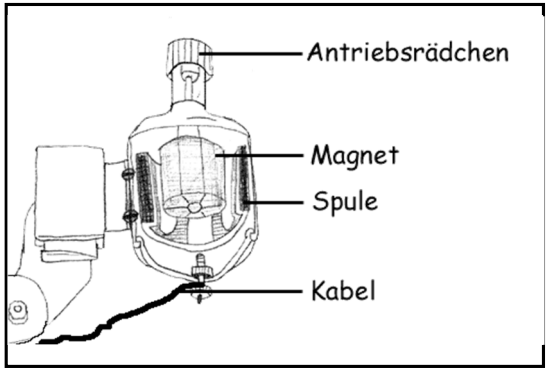


Dynamo

Induction in coil

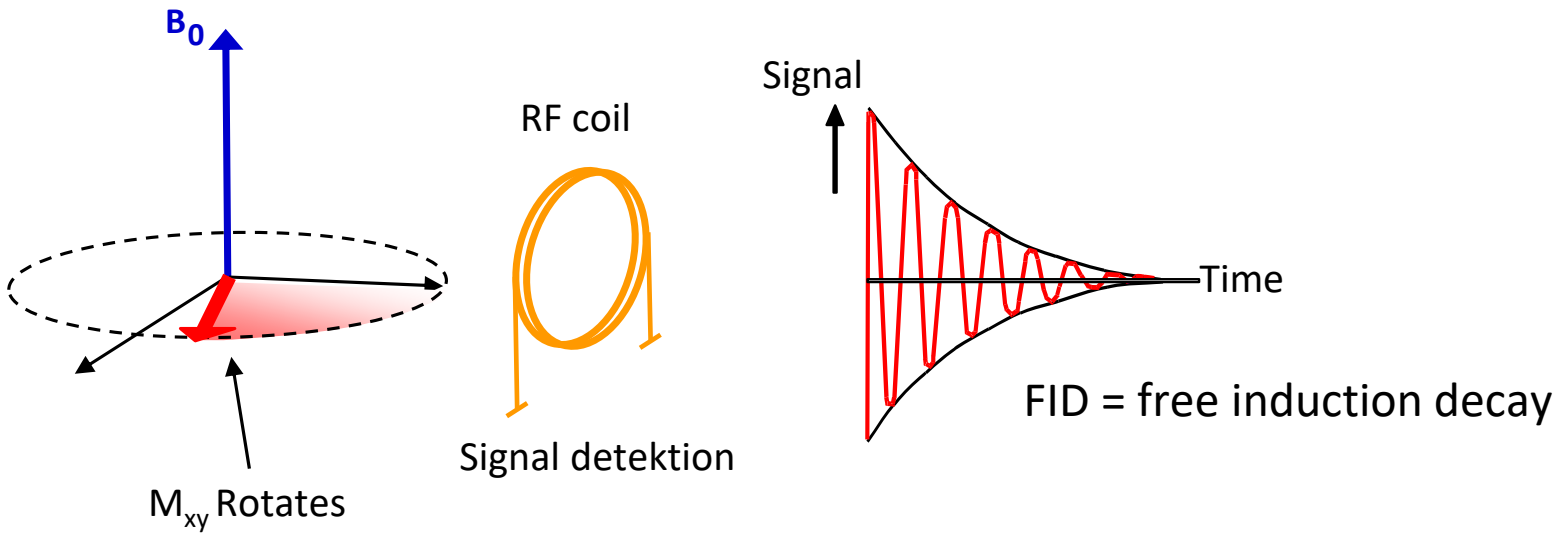






MR-Signal

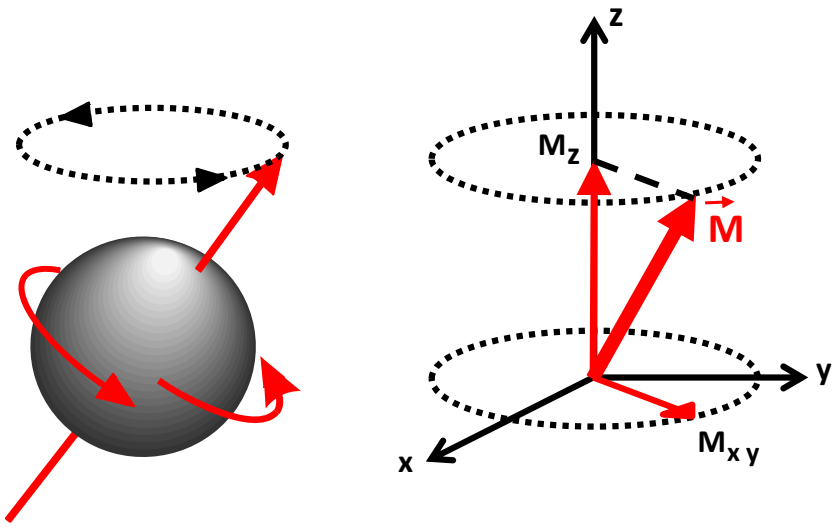
Induction in receiving coil



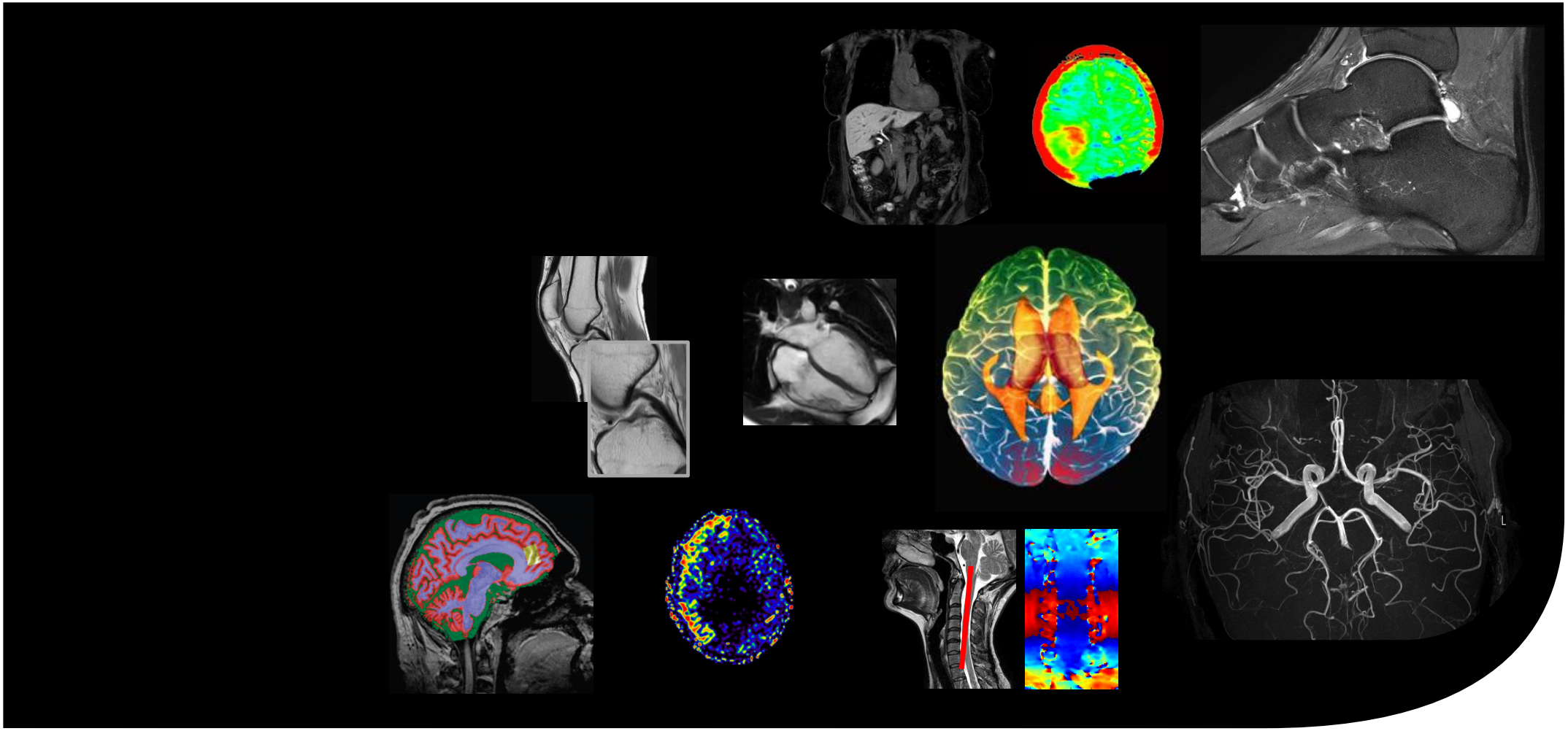
Induction in RF coil

Vector representation

Longitudinal and transverse component of magnetization



$\vec{M}$ makes precession movement

M_z longitudinal component
is static

M_{xy} transverse component
rotates around z-axis